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Evaluation of preceding variables affecting behavioral use and acceptance of chord-enabled keyboard among students

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ABSTRACT

For years, chord-enabled keyboards' usage has stayed behind its famous counterpart, the standard QWERTY design. Recent trends in this type of keyboard prompted its usage among social media platforms focusing on productivity and efficiency. However, few studies have evaluated its acceptance and consumer usage behavior – especially in the current generation. To fill this gap, this study aimed to examine the factors that affect the acceptance and adoption of chord-enabled keyboards through Random Forest Classifier and Neural Network ensemble. Nine thousand nine hundred (9,900) data points were considered through the convenience sampling method utilizing online platforms. The results displayed that all the examined latent variables were deemed significant: Performance Expectancy, Effort Expectancy, Social Influence, Hedonic Motivation, Facilitating Conditions, Price Value, Habit, Intention to Use, Usage Behavior, and Perceived Usability. Among all these, usage behavior was identified as the most influential variable affecting the acceptance of chord-enabled keyboards. Users are more likely to utilize chord-enabled keyboards due to their perception of increasing productivity in accomplishing their tasks quickly, efficiently, and comfortably in the face of multiple typing tasks. Moreover, the variables of Effort Expectancy and Facilitating Conditions were also considered significant but less influential among all latent variables – due to difficulty on adoption. The results shed light on the factors influencing chord-enabled keyboard acceptance. With this, keyboard manufacturers can utilize the results to improve the acceptance and adoption of chord-enabled keyboards to improve individuals' typing efficiency and productivity, which can be extended for promotion in different countries.

1. Introduction

Schools, colleges, and universities shifted to an online learning system due to the COVID-19 pandemic. This shift posed many challenges for teachers and students, such as adapting to new learning systems and forms of communication (Barrot et al., 2021; Biwer et al., 2021). Along with these new learning platforms, the distribution of academic requirements has also shifted from traditional printed materials to online learning materials (Kang, 2021). With the implementation of an online educational setting, schools have digitalized most outputs. Moreover, Motz et al. (2021) instigated that the transition to online learning has increased online coursework among students.

During the COVID-19 pandemic, however, this increase posed a challenge as the new online system expected students to know how to cope with fast-paced learning (Adnan & Anwar, 2020). As a result, students considered using digital software and hardware to cope with the different requirements and challenges. One of the essential hardware devices that students use for online learning is the keyboard, which allows them to submit academic output and communicate ideas (Nambiar, 2020). Hence, students require adequate keyboarding skills to handle the heavy workload. Unfortunately, keyboards vary in design – an essential factor in typing performance, which can create different advantages in terms of efficiency and accuracy (Sandnes, 2010; Swanson et al., 1997).

Among the numerous keyboard designs in the market, two of the most common are mechanical and membrane keyboards. Mechanical

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keyboards use individual switches for each key, providing a smoother and more responsive typing experience. In contrast, membrane keyboards use a rubber dome under each key for improved key press registration (Pham & Kelling, 2015). With these designs, manufacturers continuously use these standards in the highly competitive and saturated market, thus making it difficult for unconventional keyboards to rise (Kafaee et al., 2022).

Unconventional keyboard layouts and designs are rare because most prefer conventional keyboards, standardized by typewriters – the QWERTY keyboard (Zaugg & Reeve, 2021). For years, the typewriter has revolutionized work settings (Lyons, 2021). This societal impact has left a feeling of nostalgia that many still try to find today. For this reason, many consumers still prefer keyboards that mimic the typewriter's traditional layout, design, and feel (Rutherford & Shaw, 2011). However, the developer intended the typewriter's layout design to "slow down" the typist so that the type bars would not jam. Additionally, during the release of its first patent, transcribing or impressing letters was mainly aimed at legibility purposes. Although this has been incredibly beneficial in the past, typing in the current generation is mainly characterized by speed, accuracy, rendering, and comfort to keep up with the range of users, tasks, and technologies. These reasons demonstrate the desire for technological freedom from the mechanical restraints of the standardized keyboard.

The layout and design of the standard keyboard can no longer compete with the fast demand for output material in the near future. However, manufacturers have produced numerous unconventional keyboard layouts and designs to combat its limitations in recent years. In contrast to the aging decline of the traditional keyboard's output efficiency, the old yet recently trending chord-enabled keyboards offer a more efficient and ergonomic alternative. One of these featured keyboards is the DVORAK layout. This was said to be preferred during the earlier period, but the availability of QWERTY keyboard and users' influence led to the unrecognition (Cakir, 1995). Subsequently, Hwang et al. (2012) presented other types which were SSVEP-based BCI spelling system that adopts the QWERTY keyboard. However, multiple steps were needed for this to be considered as widely adaptable. Bi et al. (2010) has then presented the applicability of quasi-QWERTY keyboards. However, its applicability due to constant practice among users hindered its widespread. Therefore, continuous development and improvement of keyboards – even in present time is evident such as that of chord-enabled keyboards.

According to Gopher and Raij (1988), the chord-enabled keyboard's design allows each finger movement to send a different character through multiple keypress combinations so that minimal effort is made for the same work. Not only does this mean that a user can type quicker, but it also saves up more physical stamina for extended periods. They verified this through the results of their study, wherein users who utilized chord-enabled keyboards were almost twice as fast compared to users who utilized the QWERTY keyboard. Additionally, the mechanism behind chord-enabled keyboards is especially significant for people who are visually impaired or blind, as chord-enabled keyboards need little to no visual requirements. These keyboards may also benefit users with only one abled arm, as there are designs for one-handed chord-enabled keyboards (Rakhmetulla & Arif, 2020). Despite the evaluations of what chord-enabled keyboards can do, the acceptance and usability of these keyboards are still scarce.

Chord-enabled keyboards are still trying to become as widespread as traditional QWERTY keyboards but may appeal to only some users in the current generation. Gopher and Raij (1988) compared chord-enabled keyboard designs with the traditional QWERTY layout of keyboards. Their results showed that the chord-enabled keyboards' typing speed and efficiency were better than the standard, yet consumers still ignore better-designed data-entry devices. In relation to the current market, the keyboard industry is highly competitive (Kafaee et al., 2022). Pham and Kelling (2015) analyzed membrane and mechanical keyboards in terms of words per minute, error percentage, and surface electromyography. The study showed a significant difference between keyboard designs. However, there is no defined best keyboard design, and each layout and design have its advantages and disadvantages. In relation, the need to further consider the usability and acceptability should be examined since generational adoption, especially with tech-savvy users are lacking from recent studies. Swanson et al. (1997) further backed up the difference in keyboard capabilities as the study analyzed different layouts of keyboard and their differences in efficiency and fatigue gained from using said keyboards. That said, unconventional keyboards such as chord-enabled keyboards should also be measured on usability as they may benefit users that traditional keyboards do not provide. In addition, no current studies have evaluated this technology in its current state but are promoted in the market (Edwards, 2020).

Because Chord-Enable keyboards deviate from traditional QWERTY set-up, challenges may be seen on users' learning curve and input process (Faber et al., 2022; Tortorella et al., 2022). In addition, the study of Tortorella et al. (2022) explained that recent technological advancement may have changed the behavior of human-computer interaction. This indicates that the need for further study on the current users need evaluation so development may be done for widespread market and help in the acknowledgement of this developed hardware.

This study aimed to identify the acceptance of chord-enabled keyboards among Filipino users. Previous studies have provided sufficient evidence for the advantages of the chord-enabled keyboard against the prevailing mechanical design products. Despite the plethora of evidence that supports the superiority of chord-enabled keyboards, the records regarding their acceptance and usability online are limited. To address this gap, this study aimed to discover users' acceptance of using chord-enabled keyboards and why the standard keyboard is still widespread despite its suboptimal qualities. This aim helps to highlight unorthodox yet efficient designs and layouts of keyboards that lead to the usability and acceptance of chord-enabled keyboards.

Specifically, this study extended UTAUT2 to determine the effects of different factors on accepting the chord-enabled keyboard. This study helps to highlight the factors that affect Filipino students' preference for their keyboards. Consumers can use the result of this study to gain a better insight into chord-enabled keyboards. With the sudden shift in work and education environment, it would be better to understand the acceptance of Filipinos towards chord-enabled keyboards to provide a better design to cope with the different demands. The theories and tools utilized can also be applied and modified to analyze the acceptance of different keyboard operators in different fields of work. Managerial implications were also elaborated from the results of the study to provide insights among developers.

2. Literature review and conceptual framework

2.1. Theories

There are numerous tools utilized to measure the usability of chord-enabled keyboards. One common theory being utilized when dealing with technology acceptance or adoption would be the Technology Acceptance Model (TAM) by Davis (1989). It encompasses several variables such as perceived ease of use and perceived usability. Both of which affects attitude, behavioral intention, and acceptance. However, studies such as that of Su and Li (2021) presented that TAM has been utilized with several extended variables among related studies since the sole model lacks holistic assessment. Similarly, Deng and Yuan (2020) integrated social capital theory with TAM for holistic assessment of sustainable behavior. Especially when system characteristics are considered, usually the model may be integrated or extended for probable holistic assessment (Prasetyo et al., 2021). Their study highlighted the use of DeLone and McLean IS Success Model to holistically assess the system and its characteristics as TAM alone cannot measure this.

DeLone and McLean IS Success Model is used for assessing technology utilization. This is commonly considered for information system

evaluation among user behaviors (Aldholay et al., 2018). Under which are three variables of qualities (*i.e.,* system, information, and service) affecting user behavior. Despite being effective for overall system evaluation, the model is usually considered for system enhancement (Ojo, 2017). Thus, this is commonly a model for integration or utilized for system evaluation alone.

Other models include the Unified Theory of Acceptance and Use of Technology (UTAUT) and its development – UTAUT2. The progression of the models as per the proponents (Venkatesh et al., 2003) is only the addition of other variables. To which, the main constructs are effort and performance expectancy, social influence, and facilitating conditions. For UTAUT2, the additions are habit, hedonic motivation, and price value. The extended variables coming from established theories like TAM, Theory of Reasoned Action, and Theory of Planned Behavior provided a more concise insight for technology acceptance. This has therefore been said to be commonly used for newly developed systems or technologies as a means of overall evaluation (Ong et al., 2022).

In addition to studying usability, the study may evaluate technologies' acceptance and usage behavior using the UTAUT2. The UTAUT2 is a theory that can be used to measure the consumers' acceptance of chord-enabled keyboards (Venkatesh et al., 2003). Faqih and Jaradat (2021) utilized the UTAUT2 as a framework to assess the adoption of augmented reality (AR) in education, an emerging technology at the time that could improve home-based learning. It presented how the model could provide insights into why consumers could accept technology. However, most of the studies (Mustafa et al., 2022; Penney et al., 2021; Suo et al., 2022) were seen to disregard numerous latent variables due to their insignificance. Thus, these studies also suggested that analysts consider other latent variables to measure technology use and acceptance holistically (Gumasing et al., 2023). Their study suggested that using machine learning algorithms or tools may highlight the significant latent variables compared to traditional and multivariate statistical tools. As stated by Fan et al. (2016) and Woody (2011), the use of multivariate analyses such as structural equation modeling (SEM) has limitations such as having larger model producing low significance among variables, and the farther the variable to the target object, the lower the significance. The machine learning tools considered were the Random Forest Classifier (RFC) and Artificial Neural Network (ANN) to analyze the essential latent variables.

2.2. Machine learning algorithm

Parmar et al. (2018) describe RFC as an ensemble machine-learning technique that produces an optimum decision tree as a classification model that uses a randomly selected subset of training samples set by the user. Ensemble learning techniques such as RFC have significant applications in data mining. Analysts utilize this technique due to the accuracy of its classifications and predictions (Denisko & Hoffman, 2018). Additionally, RFC works well with large amounts of data and can handle thousands of input variables without deletion (Parmar et al., 2018). Valecha, Varma, Khare, Sachdeva, and Goyal (2018) utilized RFC to predict customer behavior that affects the choice of buying a product significantly. The study also highlighted the benefits of using RFC over other machine learning techniques due to its feasibility for a vast data set and applicability for all databases. Additionally, the study suggests this tool is an excellent aid in understanding users' thought processes on technology acceptance, such as chord-enabled keyboard. ANN was also considered for the study to further enhance the validity and accuracy of these predictions.

ANN is a computational method used for machine learning, knowledge demonstration, and maximization of the output of complex systems using knowledge gathered from learning. ANNs learn by utilizing examples provided and are used to identify patterns in each dataset during this learning process (Dastres & Soori, 2021). Muliauwan et al. (2020) compared different artificial intelligence methods in predicting concrete properties and concluded that ANN prediction results are more accurate than those from other artificial intelligence methods. Hammoudi, Moussaceb, Belebchouche, and Dahmoune (2019) also supports this conclusion as the accuracy of the ANN predictions was higher than other machine learning methods.

2.3. Conceptual framework

The conceptual framework (Fig. 1) presents the different theories affecting consumers' decisions to use and adapt to chord-enabled keyboards. It aims to determine the factors affecting consumers the most in using chord-enabled keyboards instead of the traditional design. In understanding how the different external factors would influence a person's perception of the Usability of chord-enabled keyboards, the researchers utilized a conceptual framework that extended the UTAUT2, including perceived usability which was hypothesized to affect acceptance.

Performance Expectancy (PE), a variable in the UTAUT2, demonstrates the relationship between the expected performance of the consumer towards the product. Venkatesh et al. (2003) defined PE as the degree of belief that using technology can enhance the performance of an individual. It provides an excellent reference to understand how the performance of chord-enabled keyboards and other external factors determine a user's Intention to Use (IU) the product. Some people avoid unique technology because it is foreign to them; however, this does not mean that users have no intention of using the product. Studies by Anderson et al. (2009) and Gopher and Raij (1988) regarded users' perception of using and learning chord-enabled keyboards. It stated that although the mapping of the chords poses a challenge to the keyboard's acceptance due to its high physical and cognitive demands, some are still open to its alternative keyboard design due to its better typing and overall productivity performance.

Considering this, an effort to learn the alternative keyboard also plays a vital role in the study. The difficulty of learning a new set of muscle control in using the alternative keyboard may prove challenging and serve as a deterrent. Nevertheless, ease of use is a very enticing factor that can be measured accurately by the Effort Expectancy (EE) variable of the UTAUT2. EE measures how easy it is to use information technology (Venkatesh et al., 2003). Changing from one accustomed gesture to a new, innovative one can be daunting. However, evidence suggests that using chord-enabled keyboards allowed users to accomplish a monotonic increase in typing speed despite the keyboard's physical and cognitive demands (Beddoes & Hu, 1994). Both the effort of the learning curve and the decrease in the physical effort can make the average user indecisive in their intention to use the product.

H1. Performance Expectancy has a significant direct effect on Intention to Use.

H2. Effort Expectancy has a significant direct effect on Intention of Use.

Many reasons go with deciding to use a product. One includes Social Influence (SI), the degree to which one thinks an important person influences their belief to use something new Venkatesh et al. (2003). It is worth noting SI as a primary component of the chord-enabled keyboard's popularity on social media platforms like TikTok. The platform allows its content creators to utilize it in their business, including the advertisement of chord-enabled keyboards. The combination of the content creator's marketing skills and the algorithm allowed the product to trend, increasing its popularity, which can make the users try and explore a unique alternative to the QWERTY keyboard. Kalia et al. (2022) conducted research in online retail to compile past research by multiple countries about the increase in articles published, which reflected the increase in online retailing. The research concluded that Social Influence could influence consumers' intentions. Thus, the study hypothesized that people of great importance greatly influence a user's decision to use and buy the products.

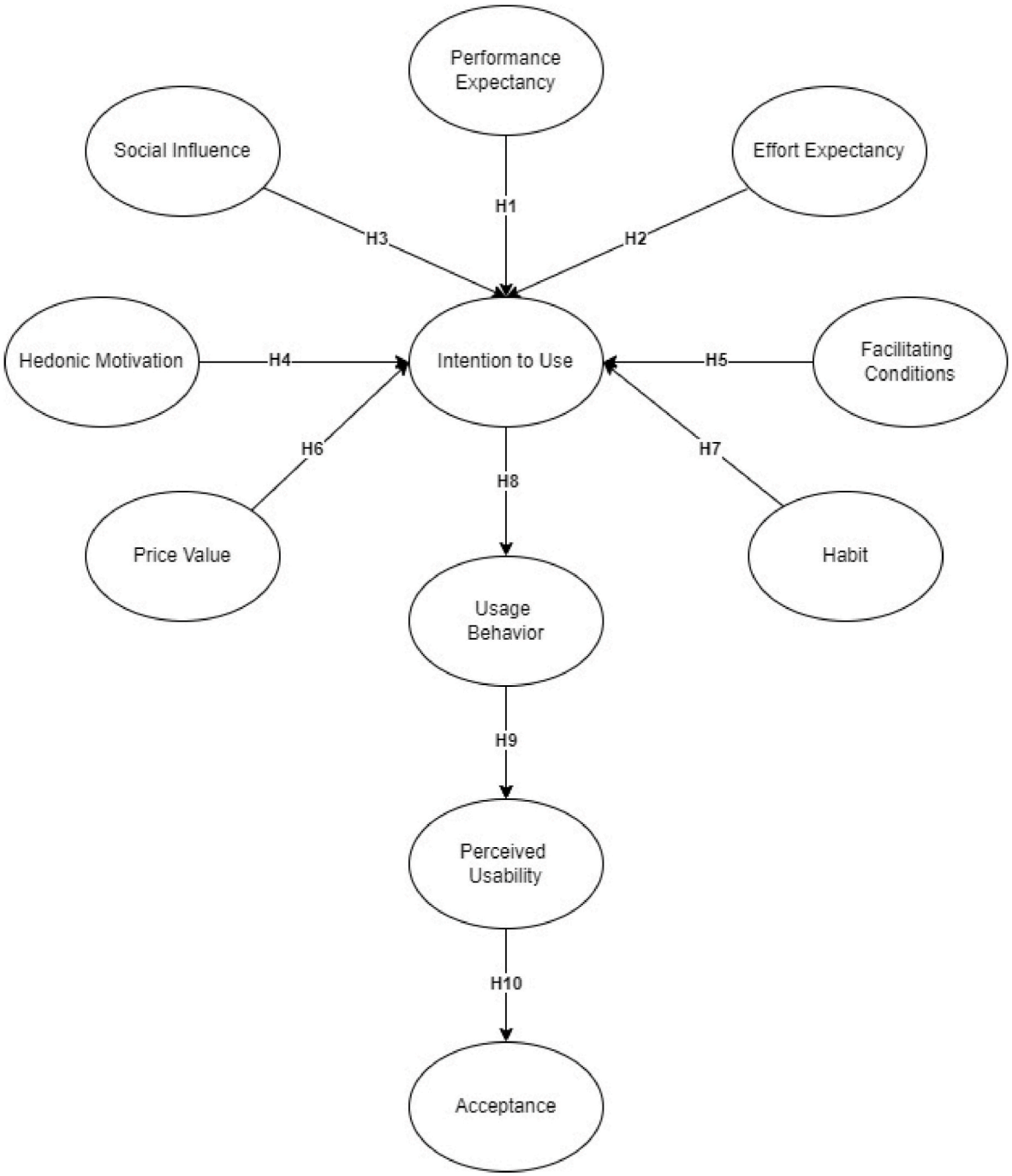


Fig. 1. Conceptual framework.

H3. Social Influence has a significant direct effect on the Intention of Use.

Hedonic Motivation (HM), another variable in the UTAUT2, describes the relationship between the emotional and experiential benefits of purchasing or using a product. Tamilmani et al. (2019) defined HM as the fun or pleasure of using technology and demonstrated hedonic motivation's significant role in determining technology acceptance and use. The study explained that the hedonic element relates to buying a product solely from the consumer's feelings rather than the product's objective qualities. A study by Bettiga et al. (2020) has shown that feelings primarily influence consumers' reactions to a new product about its utility. The study also showed that hedonic motivations are among the most critical factors in consumers' purchase of technological products. Therefore, with the variable above as highly established in technological products, it was hypothesized.

H4. Hedonic Motivation has a significant direct effect on Intention of Use.

Venkatesh (2003) refers to Facilitating Conditions (FC) as the availability of the required technical resources for the customer to support the implementation of a specific technology. She also described FC as objective factors in the environment that observers agree make an act easy. Studies by Li et al. (2019) and Tao et al. (2020) discussed consumer acceptance of health-information technologies; the studies proved that increased facilitating conditions improved users' Behavioral Intention to accept new technology. Moreover, Wang (2020) stated that having a greater level of facilitating conditions, such as the access and

availability of technical support, reduces the effort needed to use the technology when users have sufficient resources and knowledge to use said technology. Therefore, it was hypothesized that.

H5. Facilitating Conditions has a significant direct effect on Intention of Use.

Many factors are at play when a consumer plans to buy a product, and among them is the Price Value (PV) Victor et al. (2018). As such, most consumers are forced to adjust their shopping carts according to the product's price. With that said, it can be deduced that PV is important in understanding the IU of buyers. Venkatesh et al. (2003) define PV as consumers' cognitive tradeoff between the perceived benefits of the applications and the monetary cost of using them. In addition, a study by Mohamad et al. (2021) found PV to positively and significantly influence behavioral Intention to adopt technology such as mobile hotel booking. Therefore, the study has hypothesized the following.

H6. Price Value has a significant direct effect on Intention of Use.

A great deal of users are familiar with the standard QWERTY keyboard. Even jobs that use the qwerty keyboard are available, especially in office work. Each finger is assigned a letter to maximize the speed at which one can type. One should break the habit of typing with a standard keyboard to integrate the chord-enabled keyboard into society. Tamilmani et al. (2019) define a Habit (HB) as a determiner of IU. It understands the vital role of habit, in this case, typing with a standard keyboard, in the intention of a user towards a product, the chord-enabled keyboard. Introducing a new system would break all the systems the QWERTY keyboard has built. Also, habits may affect the users as they are unfamiliar with the chord-enabled keyboard. However, innovation cannot prevail as researchers have tried to understand why users are not using the chord-enabled keyboard. Thus, the hypothesis is that the user's HB dramatically influences the IU toward the product.

H7. Habit has a significant direct effect on the Intention of Use

The modified UTAUT2 integrating Perceived Usability (PU) provides a comprehensive understanding of users' acceptance of technology (Gumasing et al., 2023). A study by Nguyen et al. (2018) demonstrated how IU influences Usage Behavior (UB). The study showed that users' IU technology significantly affects their subsequent UB. This directly influences the consumer's PU, as demonstrated in a study by Albastaki (2022) and Gumasing et al. (2023). Moreover, a study by Chuenyindee et al. (2022) shows that users' PU significantly affects Acceptance. These findings show that a user's perception of PU is essential in forming their acceptance. The modified framework offers a holistic approach to understanding the relationships between IU, UB, PU, and Acceptance of Technology. For these reasons, the following were hypothesized.

H8. The intention of Use has a significant direct effect on Usage Behavior.

H9. Usage Behavior has a significant direct effect on Perceived Usability.

H10. Perceived Usability has a significant direct effect on Acceptance.

3. Methodology

3.1. Participants

The researchers conducted the study to cater to people with knowledge about keyboards. As such, the data was collected from the general populace to determine the acceptance of chord-enabled keyboards in the Philippines. That is, online survey was used among student respondents using QR codes and online social media group site dissemination in Viber, Facebook, Twitter, and Instagram. The study opted for a convenience sampling method to obtain the information needed for this quantitative study. Andrade (2020) explained how convenience sampling is better for quantitative studies regarding generalizability. Reflecting with the study of Setiawan et al. (2021), a minimum of 200 respondent was needed for the analysis. In accordance, the multivariate analysis suggestion of Hair et al. (2010) expressed that at least 150 respondent is needed for a generalized output. With approximately 500 maximum student per program per university, the Yamane Taro formula (eq. (1)) with 95% confidence showed that at least 218 sample is needed for generalization.

$$n = \frac{N}{1 + N(e^2)} \quad \text{(eq. 1)}$$

The study assessed more than the minimum, obtaining 225 valid responses from only those familiar with or know about chord-enabled keyboards. There was a total of 263 initial response and only 225 responses were considered valid and acceptance, 96.886% response rate, because of complete response and answered and they are willing to participate. The data collection process run from March until November 2023. Presented in Table 1 are the demographic profiles of the respondents which was approved by the Mapua University Ethics Committee (FM-RC-23-02-46).

From the collected participants, it could be seen that the majority of the respondents were 19 years old (33.78%), 18 years and below (25.33%), 20 years old (14.22%), and 21 years old (12.00%); either male (54.67%) or female (45.33%). Most respondents are senior high school students (75.55%) and college students (22.67%), with the remainder having a higher educational level. Lastly, most participants have an excellent familiarity with keyboards (74.22%), while some only have a very good familiarity with keyboards (19.11%), with the remainder having only fair or poor familiarity.

3.2. Questionnaire

The questionnaire utilized ten sections to evaluate latent variables such as Performance Expectancy, Effort Expectancy, Social Influence, Hedonic Motivation, Facilitating Conditions, Price Value, Habit, Usage Behavior, Intention of Use, and Perceived Usability – similar to Gumasing et al. (2023). The questionnaire used a 5-point Likert scale to assess the measures considered in this study. The questions were adapted from the related studies (supplementary file).

3.3. Data preprocessing

To ensure that the data was relevant and appropriate for the study, the data underwent pre-processing and data cleaning. Tae et al. (2019)

Table 1
Demographic profile.

Characteristics	Categories	N	%
Age	18 years old and below	57	25.33
	19 years old	76	33.78
	20 years old	32	14.22
	21 years old	27	12.00
	22 years old	14	6.220
	23 years old	12	5.340
	24 years old and above	7	3.110
Sex	Male	123	54.67
	Female	102	45.33
Educational Level	Senior High School	170	75.55
	College Student	51	22.67
	Master's Degree	2	0.890
	PhD Degree	2	0.890
Keyboard Familiarity	Poor	2	0.890
	Fair	13	5.780
	Very Good	43	19.11
	Excellent	167	74.22

explained that data cleaning is meant to lessen the influence of incorrect data on the study results. The data cleaning process utilized correlation analysis considering a correlation coefficient of 0.2 and a p-value of 0.05 to ensure the significance of the gathered data to the study as feature selection process (German et al., 2023). After the data cleaning process, the researchers aggregated the data using a mean calculation to represent the input data for the machine learning algorithm. In this study, Python Integrated Development Environment – Jupyter Notebook was considered for the random forest classifier optimization process while MATLAB was utilized for the neural network optimization (Öztürk and Başar, 2022a).

Presented in Table 2 is the descriptive statistics of the measure items, together with the skewness and kurtosis for normality. It could be posited that the dataset is normal since the normality score using Shapiro-Wilk test is within ±1.96 (Hair et al., 2021). In addition, the Harman's Single Factor Test for common method bias was conducted. The total variance obtained was 23.46%, which is with the set threshold (<50%) by Podsakoff et al. (2003).

3.4. Machine learning ensemble

The study considered the 225 by 44 data points that served as the input for the RFC parameter set to generate the optimum model. Different criteria and splitter were assessed, such as gini or entropy, and random or best – analyzed at different training and testing ratios. Moreover, the researchers considered a tree depth from 5 to 7, running each combination for 100 iterations to determine the optimum output.

Matching the set conditions of the random forest classifier, the artificial neural network preprocessing incorporated correlational analysis for data cleaning. Following data normalization, a total of 4500 runs were conducted, set with ten epochs per iteration to depict the best output from MATLAB R2021a.

Table 2
Descriptive statistics of indicators.

Factors	Mean	StDev	Kurtosis	Skewness	Normality
PE1	3.218	0.985	−0.146	0.127	−0.869
PE2	3.509	1.025	−0.232	−0.347	1.496
PE3	3.564	0.930	−0.084	−0.101	1.202
PE4	3.455	1.141	−0.63	−0.225	0.357
EE1	3.673	1.096	0.463	−0.839	−1.812
EE2	3.709	0.985	0.449	−0.667	−1.485
EE3	3.745	0.995	0.579	−0.826	−1.427
EE4	3.673	1.045	0.453	−0.770	−1.700
SI1	3.018	1.000	−0.656	0.187	−0.285
SI2	3.400	1.185	−0.417	−0.294	0.705
SI3	2.836	1.187	−0.787	0.260	−0.330
SI4	2.855	1.135	−0.896	0.142	−0.158
PV1	2.964	1.061	−0.685	0.074	−0.108
PV2	2.927	0.970	−0.286	0.396	−1.385
PV3	3.018	0.963	−0.627	0.340	−0.542
PV4	2.909	1.066	−0.701	0.187	−0.267
HM1	3.291	0.985	−0.486	0.197	−0.405
HM2	3.764	1.044	−0.509	−0.490	0.963
HM3	3.000	1.062	−0.505	0.094	−0.186
HM4	3.491	0.970	−0.422	−0.158	0.374
FC1	3.873	1.063	0.936	−1.047	−1.118
FC2	3.509	1.189	−0.566	−0.556	0.982
FC3	3.345	1.048	−0.251	−0.258	1.028
FC4	3.509	1.110	−0.067	−0.121	1.806
HB1	3.400	1.215	−0.866	−0.323	0.373
HB2	3.600	1.072	−0.369	−0.495	1.341
HB3	3.218	1.056	−0.996	0.115	−0.115
HB4	4.509	6.666	0.702	1.187	1.691
IU1	3.382	1.087	−1.019	−0.036	0.035
IU2	3.400	1.072	−0.659	−0.233	0.353
IU3	3.364	1.085	−0.455	−0.253	0.556
IU4	3.491	1.093	−0.656	−0.405	0.617
PU1	3.600	0.965	−0.269	−0.355	1.320
PU2	3.491	1.110	−1.355	−0.017	0.012
PU3	3.509	1.007	−0.595	−0.190	0.319
PU4	3.545	0.969	−0.354	−0.317	0.895
UB1	3.545	1.059	−0.787	−0.312	0.396
UB2	3.382	1.087	−0.757	−0.211	0.279
UB3	3.582	0.985	−0.396	−0.352	0.889
UB4	3.418	1.155	−0.751	−0.231	0.307
AU1	3.491	1.042	−0.775	−0.223	0.288
AU2	3.455	1.041	−0.404	−0.223	0.552
AU3	3.618	1.070	−0.313	−0.546	1.744
AU4	3.527	1.076	−0.889	−0.208	0.234
AU5	3.564	0.890	−0.722	0.041	−0.057

4. Results

4.1. Random forest classifier

The RFC output considered the optimum depth at 6, with gini as the criterion and best as the splitter. The data show that the training and testing ratios 90:10 presented the best output. Since this is the only parameter combination with consistent output, as seen in the standard deviation results, the researchers considered it to generate the optimum tree for classifying acceptance and use of the chord-enabled keyboard. Presented in Table 3 are the average output and standard deviation at tree depth 6.

The results provided by the RFC show insights into the determinants of acceptance of chord-enabled keyboards. Fig. 2 presents the decision tree model, with Usage Behavior (X_1), the parent node, considered the most crucial factor. If the parent node's value is less than or equal to 0.402, it will consider X_1 with a value less than or equal to −0.984. Then, it will consider Price Value (X_0) with a value less than or equal to −2.839. Satisfying this will indicate a very high acceptance of chord-enabled keyboards; otherwise, it will consider X_0 with a value of less than or equal to 3.018 and X_0 and X_1, leading to very high acceptance. If the first child node X_1 is unsatisfied, it will consider X_1, leading to Perceived Usability (X_4). On the one hand, failing to satisfy the child node X_4 with a value less than 2.029 will result in a high acceptance of the chord-enabled keyboard.

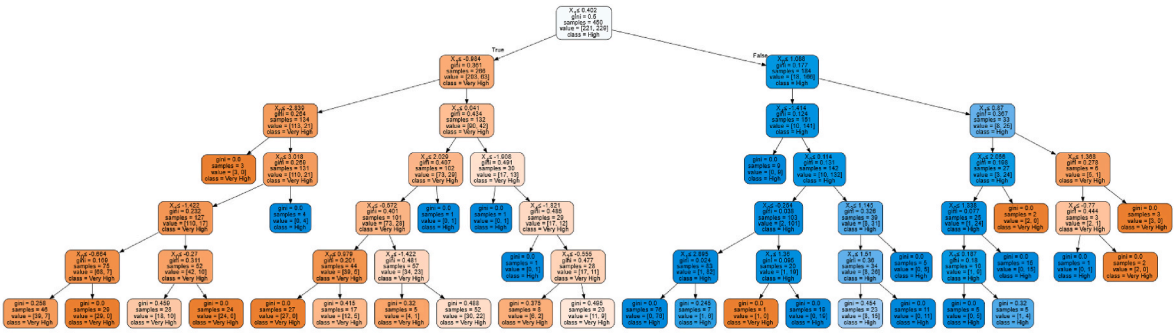
Considering the dissatisfaction of the parent node, X_0 will be considered with a value less than or equal to 1.088. Satisfying this condition will consider X_4 and X_0, leading to high acceptance. Otherwise, it will consider the intention to use X_3 with a value less than or equal to 1.145, leading to a high acceptance of chord-enabled keyboards. However, should the value fail to satisfy X_0 with a value of less than or equal to 1.088, it will consider X_1, X_0, and Usage Behavior (X_2), leading to a high acceptance towards chord-enabled keyboards.

4.2. Artificial neural network

The implementation of optimization process utilizing Levenberg-Marquardt Algorithm, low MSE values were obtained from training (80%) – 0.000170258, validation (10%) – 0.05832027, and testing (10%) – 0.0878268. In accordance, high r-squared values were obtained

Table 3
Random forest classifier summarized results.

Category	60:40	70:30	80:20	90:10
Best				
Entropy	84.38	83.52	84.49	90.92
Standard Deviation	1.311	0.944	0.802	1.002
Gini	87.52	82.74	88.49	92.00
Standard Deviation	1.245	0.704	0.502	0.000
Random				
Entropy	82.95	82.02	85.38	86.14
Standard Deviation	4.938	3.983	4.779	5.304
Gini	83.80	80.58	84.42	85.36
Standard Deviation	3.936	3.750	3.798	5.981



Legend:

X_0 – Price Value
X_1 – Usage Behavior
X_2 – Hedonic Motivation
X_3 – Intention to Use
X_4 – Perceived Usability

Fig. 2. Optimum RFC classification Model
Legend:
X_0 – price Value
X_1 – usage Behavior
X_2 – hedonic Motivation
X_3 – intention to Use
X_4 – perceived usability.

as seen in Fig. 3, where Öztürk and Başar (2022a) explained to be highly acceptable (r-squared >0.90). Presented in Fig. 4 is the optimum neural network model.

The SHAP package was then utilized to generate the score of importance to validate the findings (German et al., 2023; Ong, 2022). Table 4 presents the normalized score of importance, showing consistency with the optimum tree output. Usage behavior (UB) showed the highest importance, followed by price value (PV), hedonic motivation (HM), Intention to use (IU), and perceived usability (PU), which also showed a high significance effect on acceptance. Other factors such as habit (HB), social influence (SI), performance expectancy (PE), effort expectancy (EE), and facilitating conditions (FC) were also deemed significant.

Further validation was conducted as this newly established method has yet been widely considered among human factors-related studies. Presented in Table 5 is the path analysis summarized output for justification using partial least square structural equation modeling (PLS-SEM) algorithm. With individual hypotheses being tested, similar output is seen. That is, all relationship were deemed significant and UB, PV, HM, IU, PU, and HB were the highest coefficients – relating to higher effect on acceptance of chord-enabled keyboards. It could be seen that IU and PU were relatively close. In addition, lower coefficients were seen on SI and PE; least were seen on both EE and FC. As explained by Gabriel et al. (2018), t-statistics when using a two-tailed test should be greater than 1.96 and p-value should be less than 0.05 for it to be accepted. In the case of this study, it could be seen that all were significant. Moreover, the internal validity (convergent validity) and discriminant validity were seen to be within the threshold, all of which are presented in the supplementary material. The model fit presented that the standardized root mean square residual (SRMR) resulted to a value of 0.06, chi-square value of 2.970, and 0.904 on the normal fit index – all of which are within the suggested threshold of less than 0.08 and 5.00, and greater than 0.9, respectively (Baumgartner & Homburg, 1996; Hooper et al., 2008; Hu & Bentler, 1998).

5. Discussion

The RFC presented the viability and accuracy of the developed models. Having a 92% accuracy rate, it was further validated using the score of normalization with the SHAP package – similar with the high results of ANN. To this, UB was seen to be the most significant variable among the factors influencing acceptance of chord-enabled keyboards (Hypothesis 9). Users are more likely to utilize chord-enabled keyboards to increase their productivity in accomplishing their tasks quickly, efficiently, and comfortably in the face of multiple typing tasks. This finding aligns with the study by Beddoes and Hu (1994). It could also be posited how a strong influence of UB on actual use may be due to consumers prioritizing the practicality and performance improvements of chord-enabled keyboards. This is similar to the study of Lin and Yu (2023). However, despite the practical benefits of chord-enabled keyboards, common chord-enabled keyboards are expensive, making some users keen on measuring their finances before purchasing one.

This represents why PV affected chord-enabled keyboard acceptance significantly (Hypothesis 6). The participants perceived the chord-enabled keyboard as an affordable investment, influenced by its product value if they get the efficiency and productivity value out of it. However, technology, especially new to the users, may take some time to get used to. This study showed that perception among users regarding price may be sensitive, but consumers highly value the gain in typing performance and efficiency. This result aligns with a study by Venkatesh et al. (2012) stating that the correct price setting of a product influences the user's perception of quality, which can be applied to how the pricing of chord-enabled keyboards can significantly influence users' perception of its value. It is worth noting that the current price of the trending chord-enabled keyboard, the Characorder One, runs for around $300. With this, manufacturers should consider pricing strategies to lessen this expense and enhance the keyboard's availability and acceptance over a broader consumer base.

Besides efficiency, consumers also consider emotional and enjoyment factors when it comes to purchasing Tamilmani et al. (2019), so another marketing strategy that manufacturers should employ besides its prices and practical benefits is the enjoyment factor that comes with

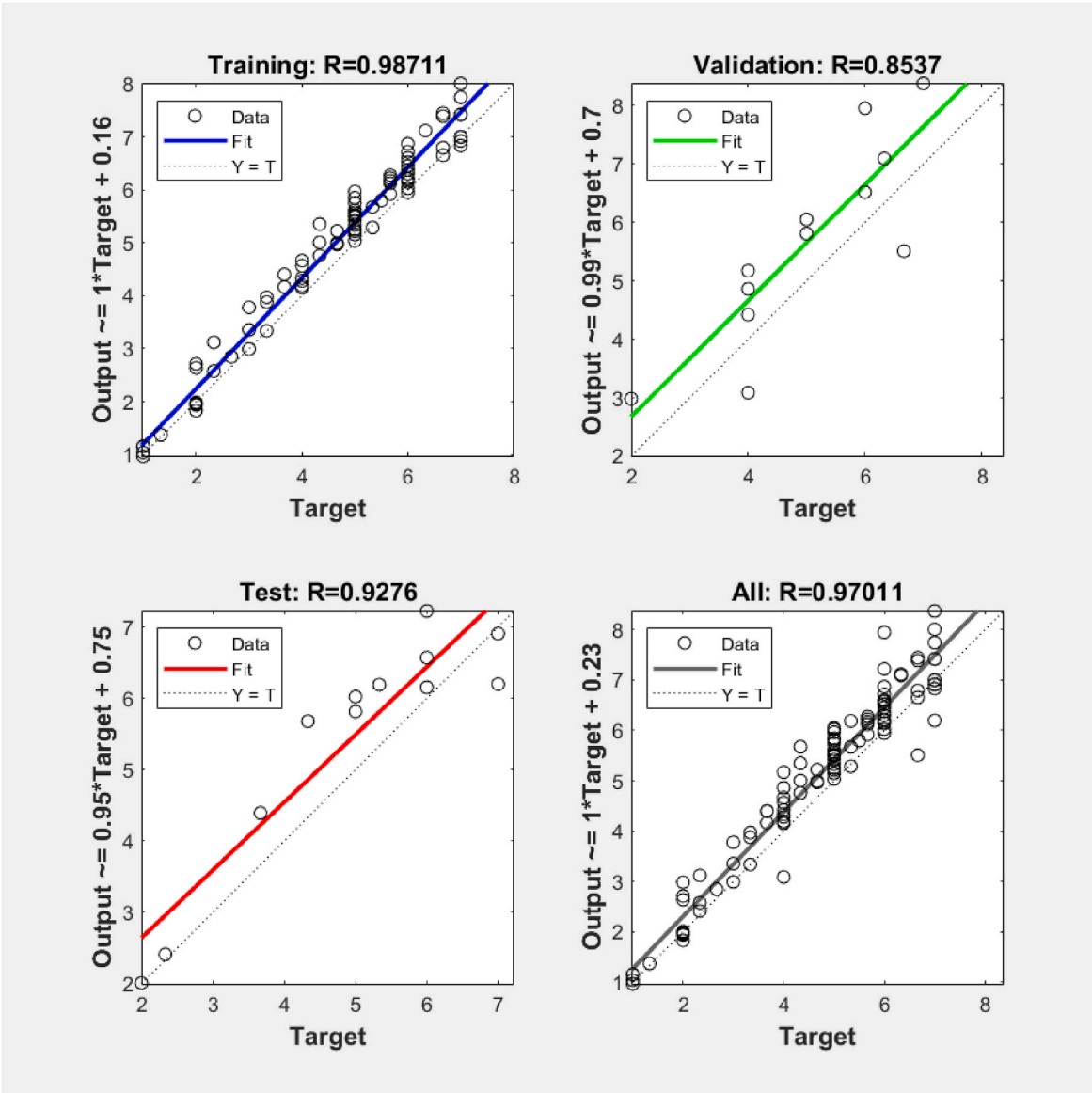


Fig. 3. ANN R-squared output.

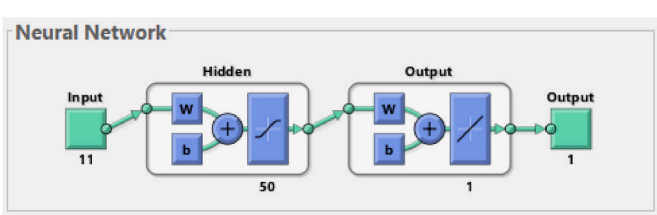


Fig. 4. ANN Optimum Model (50 nodes).

chord-enabled keyboards. This presents why HM emerged as the third most significant variable influencing acceptance (Hypothesis 4). Venkatesh et al. (2012) asserted the importance of hedonic factors in influencing technology acceptance. However, research by Aburbeian, Owda, and Owda (2022) demonstrated that self-efficiency and effectiveness also strongly influenced acceptance, indicating that consumers might prioritize practicality over hedonic factors. However, beyond the barrier that differentiates these motivators, the consumers are proven to have their intentions as to why they will use them upon having positive measure items.

Table 4
Normalized score of importance.

Latent Variable	Importance	Normalized Importance
UB	0.140	100
PV	0.138	98.6
HM	0.133	95.2
IU	0.131	93.7
PU	0.129	92.4
HB	0.128	91.0
SI	0.124	88.3
PE	0.124	88.3
EE	0.105	75.2
FC	0.104	74.5

IU is another latent variable deemed significant to the acceptance and future usage of chord-enabled keyboards (Hypothesis 8). Seeing essential measure items from prior significant latent variables, consumers agreed that the efficiency and productivity of output using the chord-enabled keyboard would significantly influence their acceptance

Table 5
Path analysis summary.

Hypothesis	Relationship	Coefficient	T Statistics	P Values	Decision
1	PE → IU	0.204	2.028	0.018	Accept
2	EE → IU	0.173	2.553	0.021	Accept
3	SI → IU	0.513	3.401	0.004	Accept
4	HM → IU	0.803	5.362	0.001	Accept
5	FC → IU	0.146	2.233	0.032	Accept
6	PV → IU	0.886	5.262	0.001	Accept
7	HB → IU	0.649	6.512	0.009	Accept
8	IU → UB	0.794	8.204	<0.001	Accept
9	UB → PU	0.909	8.523	<0.001	Accept
10	PU → AC	0.797	6.875	<0.001	Accept

and usage behavior. From the measure items, it was seen that people are willing to use it in the future, open to its purchase when it becomes accessible, and would even recommend it. Gumasing et al. (2023) explain that when consumers perceive great benefits in technology, a high intention will be seen in its acceptance and usage. It also mentioned that when technology helps students in their academic tasks, greater intention is seen on them using it. Thus, their PU plays a significant role in acceptance.

In the formation of usage plans, PU emerged as a significant variable (Hypothesis 10). The results indicated that participants perceived chord-enabled keyboards as likable, well-integrated, and easy to learn and use. The study by Albastaki (2022) and Barrot et al. (2021) examined PU in the context of numerous technologies and its implications on acceptance. The user-friendly nature of chord-enabled keyboards contributes to their PU, subsequently influencing acceptance. Through its user-friendly nature and positive PU, the possible adoption of repeated usage among chord-enabled keyboards may be high, indicating a high possibility of users forming habits and integrating the keyboard into their daily lives.

The data shows how HB is a significant latent variable in users' acceptance of chord-enabled keyboards (Hypothesis 7). Participants showed positive behavior toward incorporating chord-enabled keyboards into their daily routines and utilizing them as a natural part of their day may be done effortlessly. This observation aligns with the habit formation concept that explores its role in users' acceptance of technologies (Alabastaki, 2022; Barrot et al., 2021). Thus, the habitual use of chord-enabled keyboards reinforces acceptance and future usage. However, habitual usage often influences the social environment a user is immersed in (Carden & Wood, 2018). In this case, the environment of the user mirrors the desired action. For this reason, social factors are also deemed influential in the formation and acceptance of chord-enabled keyboards.

SI is another significant variable deemed to influence chord-enabled keyboard acceptance (Hypothesis 3). The result indicated that the opinions and behaviors of essential people in the participants' lives, such as friends, colleagues, and peers, impacted their inclination toward chord-enabled keyboard usage. They expressed a desire to use it when they see others using it. The finding parallels the concept of social influence, which demonstrates that social perceptions influence individuals' actions (Venkatesh et al., 2012). The influence of the usage and ownership of family and friends on an individual's purchase of input hardware was also discussed by Mohamad et al. (2021), who examined the role of SI in shaping technology acceptance. Their study suggested that SI is highly significant in shaping the user's acceptance and adoption. Besides the role that peers and families play in accepting this new technology, the intrapersonal perception of performance and effort are still considered in chord-enabled keyboards among users.

PE and EE emerged as significant variables (Hypotheses 1 and 2, respectively). The participants showed positive expectations with the performance benefits of chord-enabled keyboards, believing that using this keyboard type would enhance their productivity and efficiency in accomplishing typing tasks. These results align with performance expectancy, where individuals would focus on a technology's potential to improve task performance. The participants also expressed positive expectations regarding the effort required to use chord-enabled keyboards. They believed that the learning curve of this keyboard type would be fast and easy. These findings align with effort expectancy, wherein users will perceive technology as easy to learn. In line with this, a previous study by Wang et al. (2020) also highlighted the importance of both variables in forming an individual's acceptance and adoption of chord-enabled keyboards. Binyamin and Hoque (2020) explained that technology's ease of use and usefulness affect consumers' acceptance – especially in education (Zheng & Li, 2020).

FC was also seen as a significant but less influential variable (Hypothesis 5). Participants expressed their perception of having access to the necessary resources and assistance when facing challenges in utilizing chord-enabled keyboards. They also expressed the keyboard's compatibility with their work-setup. While it ranked the lowest among all latent variables, it still shapes the individual's acceptance and adoption of the keyboard. The study by Venkatesh et al. (2003) deemed the significance of FC in the acceptance and adoption of technology. However, it is worth noting that most individuals today often default to using the standard keyboard due to its general availability and familiarity. Therefore, manufacturers and online resource platforms are influential in providing a comprehensive and supportive environment that encourages the use and adoption of chord-enabled keyboards.

From the overall findings, the results shed light on the factors influencing chord-enabled keyboard acceptance. The results demonstrated that all these factors significantly impacted the keyboard's adoption. These results provide valuable insight for manufacturers, marketers, and educators to promote better the adoption and overall user experience of the chord-enabled keyboard. Nonetheless, further research is needed to explore the relationship between these factors and address barriers to acceptance, such as PV and FC. By doing so, individuals can foster a wider acceptance of the device among Filipinos and jumpstart its integration into daily typing practices. Despite this, participants deemed the keyboard acceptable, with its high potential for efficiency and less ergonomic side effects. It was also highlighted that users may have challenges using the keyboard but are adaptable and willing to learn how to use the chord-enabled keyboard.

5.1. Theoretical implications

The study extended the UTAUT2 framework to determine the behavioral intention of Filipinos to utilize chord-enabled keyboards. Significant factors, such as Usage Behavior, Price Value, Hedonic Motivation, Intention of Use, Perceived Usability, and Habit, immensely influence their decision to accept the chord-enabled keyboard. The data in Table 3 infers that targeting the Social Influence, Performance Expectancy, Effort Expectancy, and Facilitating Conditions factors may further increase the acceptance of the chord-enabled keyboard in the market. Should information and awareness spread about the chord-enabled keyboard, it would feel less daunting to the people, and they may even feel encouraged to try – which manufacturers may consider and capitalized on. Utilizing the chord-enabled properties and benefits can induce a profit for the manufacturers. Users are inclined to use innovative and new technology when the opportunity arises with its low price and interesting features and benefits, as seen in the results. Thus, other researchers may consider the extended UTAUT2 utilized in this study to holistically evaluate use intention, perceived usability, and, eventually, technology acceptance – even for newly developed technology and systems. Specifically, intention was seen to affect usage behavior and perceived usability. Moreover, the utility of system usability scale measure item and rescaling the items to fit the specific system or technology was seen to be beneficial.

In another context, most studies have focused on the use of multivariate tools such as structural equation modeling. With many disadvantages being deciphered by related studies (Fan et al., 2016; Woody,

2011), the application of machine learning algorithm as classification tools were deemed to be advantageous. That is, this study provided a high accuracy of 92% for RFC and 97% for ANN. This means that future research may opt to apply other machine learning techniques, even higher power algorithm for prediction like long short-term memory (LSTM). This study paved a way for other techniques to be applied.

5.2. Managerial insights and practical implications

Chord-enabled keyboards are those designed to provide a faster and more efficient device to input characters into the computer. With its simple yet effective design, it can combat an average keyboard in terms of different factors such as portability, ergonomics, accessibility, and typing speed. Interestingly, the competition among various keyboard layouts and designs is fierce (Anderson et al., 2009). Despite the evidence and research that suggest how chord-enabled keyboards surpass the average keyboards in the market, these companies still somehow limit their products to mechanical and membrane keyboards. Using different social media platforms and advertisements among these methods may carry variables such as HB, SI, PE, EE, and FC. Gaining the fast-paced marketability and profitability of the trending chord-enabled keyboard in social media, the company can have a small moment of monopoly in the market. It is therefore suggested that developers and companies can consider current marketing trends and strategies like prompting the use of chord-enabled keyboards among influencers and celebrities in social media to gain momentum in the market. With the current influence social media has, more younger generation may opt to consider learning and adopting to the keyboard. As seen, most of the respondents are of the younger generation, which can easily grasp new concepts and adaptation to developed technology. As suggested by Lina et al. (2022), with the development of e-marketing and heavy influence of social media personality, generation Z and younger population may opt to easily purchase trendy gadgets and technology as priced reasonably. Therefore, developers may strategize on this aspect to promote the utility of chord-enabled keyboards.

5.3. Limitations

Despite the valuable findings from the results and thorough study, this study still considers several limitations. First, the study only focused on acceptance perception of chord-enabled keyboards among Filipinos. Future studies may consider studying other target demographics as their respondents. Extending their research to other products – such as comparison of use, comfort, and efficiency may also be a viable option for future researchers. Second, the study mostly measured the acceptance of students as collected samples. It could be improved by analyzing the difference in acceptance by the varying age ranges of the users, purpose, and even applicability. This would open other factors that the researchers may have missed, even implications that manufacturers may consider. In addition, considering the occupation of the respondents may also be analyzed as an extended factor covering hidden aspects such as the time used in their work environment, time allowance to learn, and budget for the chord-enabled keyboard. The determination of this response may be utilized to correlate future research findings, create market segmentation, and even fully evaluate the product in behavioral analysis, physical, and use aspects. Moreover, physical ergonomic study may be applied and determined to compare and contrast the chord-enabled keyboard with the other keyboards for better strategy development. Lastly, only RFC and ANN were considered in this study. Further analysis using other tools such as clustering techniques, may be considered by researchers to holistically evaluate chord-enabled keyboards – even in different countries.

6. Conclusion

Due to the COVID-19 pandemic, as schools shifted to online learning, academic requirements must be done digitally. With this shift to online learning and digital outputs, software and hardware are needed to be utilized. As such, one of the essential hardware needed for online learning is the keyboard. The standard QWERTY keyboard's layout and design may not compete with the fast demand for output material. With this, some manufacturers have created and further developed an unconventional keyboard designs and layouts. Chord-enabled keyboards, which has been available and has recently trend in social media, can combat this problem using a more efficient and ergonomic design with highly efficient output. This study used machine learning techniques to assess the acceptance and use of chord-enabled keyboards.

Utilizing both Random Forest Classifier (RFC) and Artificial Neural Network (ANN), it was seen that UB, PV, HM, IU, and PU significantly affected the acceptance of chord-enabled keyboards in the market. It was presented that the significance of these variables was mainly driven by the productivity, comfort, and entertainment benefits that the chord-enabled keyboard offers. This created positive perceptions of the keyboard's value. Combining these positive perceptions with the keyboard's ease of use and the consumer's intentional usage plans has significantly increased its use behavior intention and adoption. Interestingly, EE and FC had a lower influence in shaping consumers' adoption of chord-enabled keyboards. For one, the standard QWERTY keyboard's familiarity has proven to be a barrier for users to switch to a chord-enabled keyboard. Additionally, its steep learning and few assisting resources concerning the challenges of using chord-enabled keyboards have only increased the effort to make this switch.

With the findings, manufacturers can develop and rehash the keyboard and its features to further improve the FC surrounding the challenges of using chord-enabled keyboards and lessen the effort. Additionally, strengthening the first five significant variables can further encourage consumers to accept chord-enabled keyboards and induce a profit for the manufacturers. The theories and tools utilized can also be applied and modified to analyze the acceptance of different keyboard operators in different fields of work. Moreover, the results and application of this study can be applied and extended among other hardware worldwide.

CRediT authorship contribution statement

Ardvin Kester S. Ong: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Calil C. Aceron:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Warrick Jathniel S. Quimpo:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Derek Tyler U. Ong:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **John Francis T. Diaz:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Josephine D. German:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chbr.2024.100482.

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